

6	Two metal plates X and Y are contained in an evacuated container and are connected as shown in Fig. 6.1. Metal plate X is then illuminated with monochromatic light.
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[Turn over

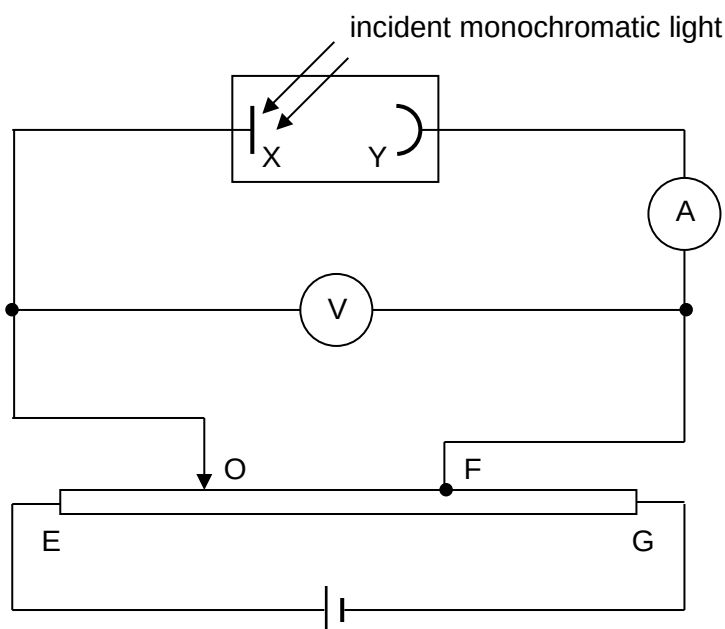


Fig. 6.1

The graph shown in Fig. 6.2 depicts the relationship between the voltmeter reading V and the ammeter reading I .

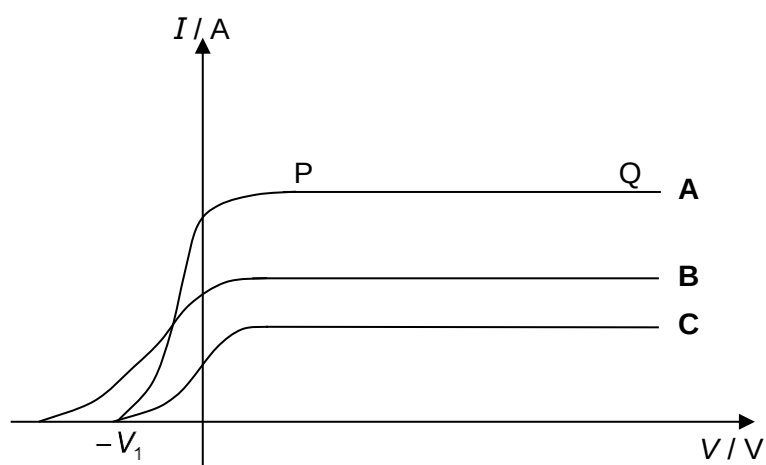


Fig. 6.2 (not to scale)

- (a) A student obtained the part PQ on graph A by shifting the sliding contact O.

State and explain where the position of O along EG should be shifted to for the student to obtain part PQ of graph A.

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			[4]
L2	Part PQ represents the saturation/maximum current. <u>At the current position of O, the potential of X is higher than Y which will create an electric field between them directed towards plate Y.</u> <u>Photoelectrons/electrons are negatively charged and thus will experience an electric force towards plate X which retards their motion.</u> Depending on where O is between E and F, <u>only some or none of the photoelectrons have sufficient kinetic energy to reach plate Y. Hence current measured in the circuit will not be maximum in value.</u> Therefore, <u>for saturation/maximum current to be measured, O would need to be shifted in the region FG / between F and G / to the right of F.</u> <u>This allows for Y to be at higher potential than X. The electric force experienced by the photoelectrons would then act towards Y and accelerate all of the photoelectrons towards Y.</u>	B1 B1 B1 B1	
(b)	Given that the work function energy of X is 1.3 eV and the wavelength of the light is 550 nm, calculate the value of the stopping potential V_1 .		
		$V_1 = \dots\dots\dots$ V	[2]
L1	$hf = \Phi + KE_{MAX}$ $\frac{hc}{\lambda} = \Phi + eV_s$ $\frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{550 \times 10^{-9}} = (1.3)(1.60 \times 10^{-19}) + (1.60 \times 10^{-19})V_s$ $V_1 = V_s = 0.96023 = \mathbf{0.96\ V}$	M1 A1	
(c)	Describe the changes, if any, to the intensity and frequency of the incident monochromatic light that the student made to obtain graphs B and C if the same metal plate X is used.		
	graph B:		
	graph C:		

			[2]
	L2	Graph B has a smaller maximum current and a larger stopping potential than graph A. The larger stopping potential indicates that an <u>incident light of higher frequency (or shorter wavelength) is used</u>. Intensity of the light is unchanged or lesser or greater (but increase in Intensity not as much as increase in frequency). Graph C has a smaller maximum current but has the same stopping potential as graph A. Since stopping potential is the same, the <u>frequency (or wavelength) of the incident light must be unchanged</u>. The smaller maximum current is due to a <u>lower intensity of light used</u>.	B1 B1

[Total: 8]

7	(a)	The decay of radioactive nuclei is said to be <i>random</i> and <i>spontaneous</i>
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